Elastic Load Balancer.

What ELB actually does

ELB automatically **distributes incoming traffic across multiple targets** (EC2, containers, IPs) so your app:

- doesn't crash under load
- stays highly available
- scales without manual intervention

If you're not using a load balancer in production, you're asking for downtime. Simple.

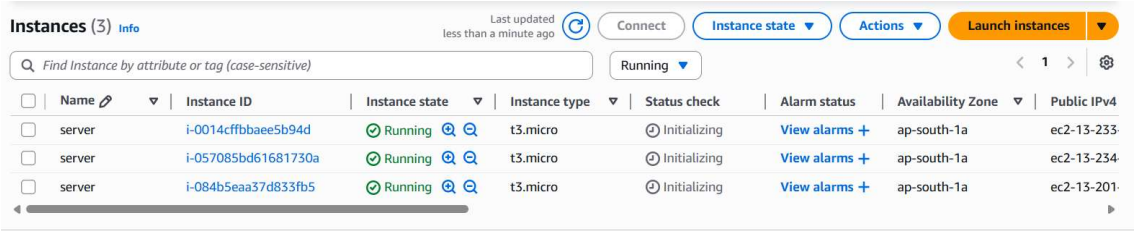
Types of Load Balancers.

- **Application Load Balancer (ALB):** This operates at the application layer (Layer 7 of the OSI model) and is ideal for HTTP and HTTPS traffic.
    - **Key Features:** It provides advanced, content-based routing rules (e.g., based on URL path, host headers, HTTP methods, or query string parameters) to direct requests to different backend services.
    - **Best For:** Microservices architectures, container-based applications, and general web applications.
- **Network Load Balancer (NLB):** This operates at the transport layer (Layer 4) and is best suited for high-performance TCP, UDP, and TLS traffic.
    - **Key Features:** It offers extremely high performance, very low latency (~100 microseconds), and the ability to use static IP addresses. It also preserves the source IP of the client.
    - **Best For:** Extreme performance scenarios, workloads requiring static IPs, or non-HTTP/S protocols.
- **Gateway Load Balancer (GWLB):** This operates at the network layer (Layer 3) and is designed for deploying, scaling, and managing third-party virtual appliances such as firewalls, intrusion detection systems, and other security or networking appliances.
    - **Key Features:** It uses the GENEVE protocol to exchange application traffic and operates transparently within the network path.
    - **Best For:** Centralized inspection and management of network traffic using virtual appliances.

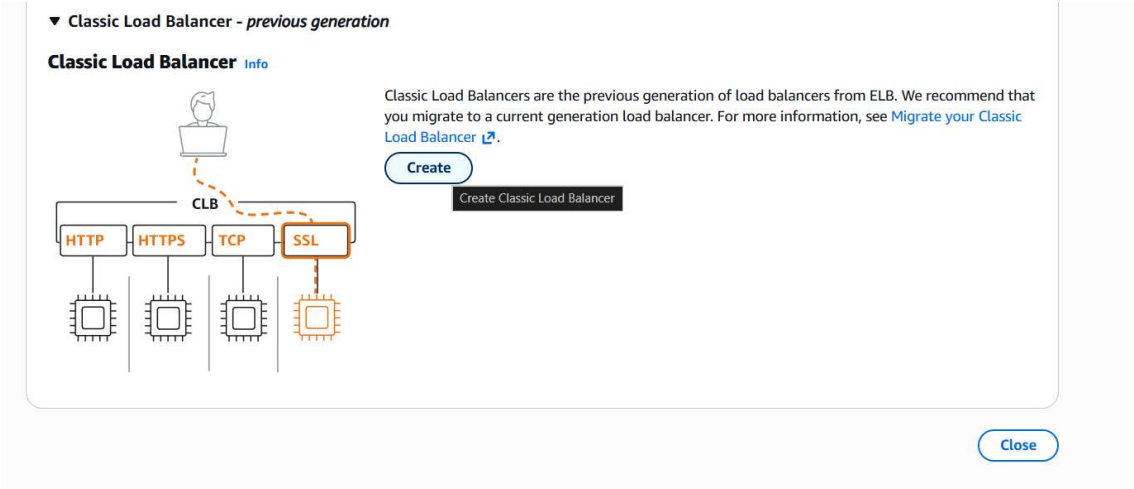
- **Classic Load Balancer (CLB):** This is the previous generation of load balancers, still available but not recommended for new applications.
    - **Key Features:** It operates at both Layer 4 and Layer 7, providing basic load balancing across multiple EC2 instances. AWS encourages migration to ALB or NLB for enhanced capabilities.
    - **Best For:** Legacy applications that are still running on the EC2-Classic network.

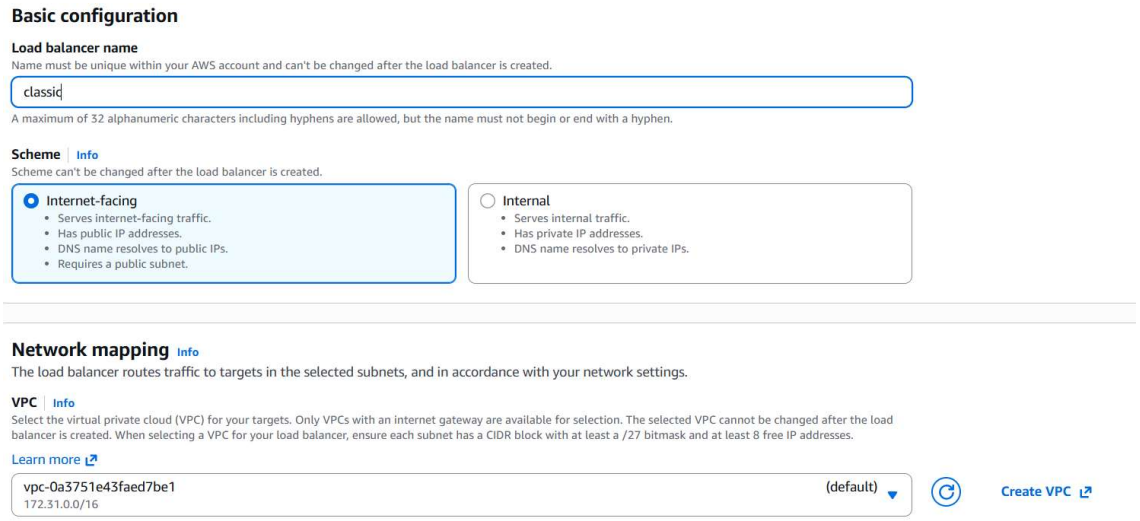
I have created the Classic Load balancer with three instances.

Three Created EC2 Instances.

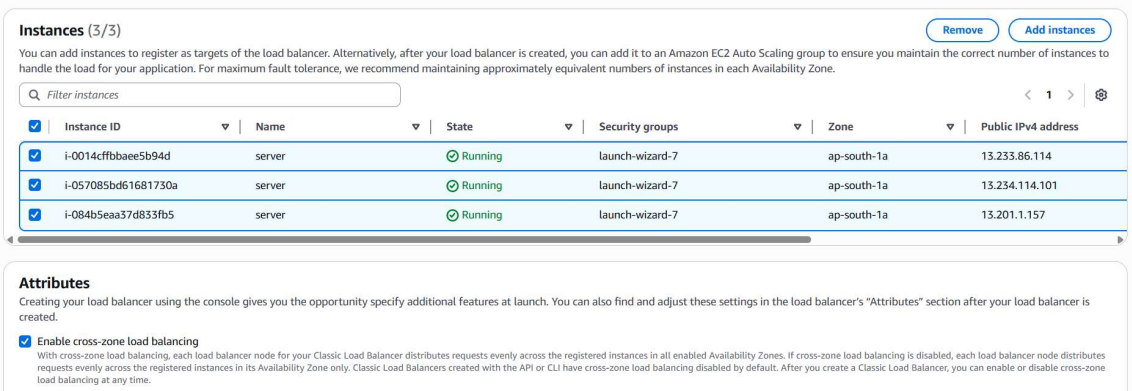


Configuration of Elastic Load Balancer – Classic .

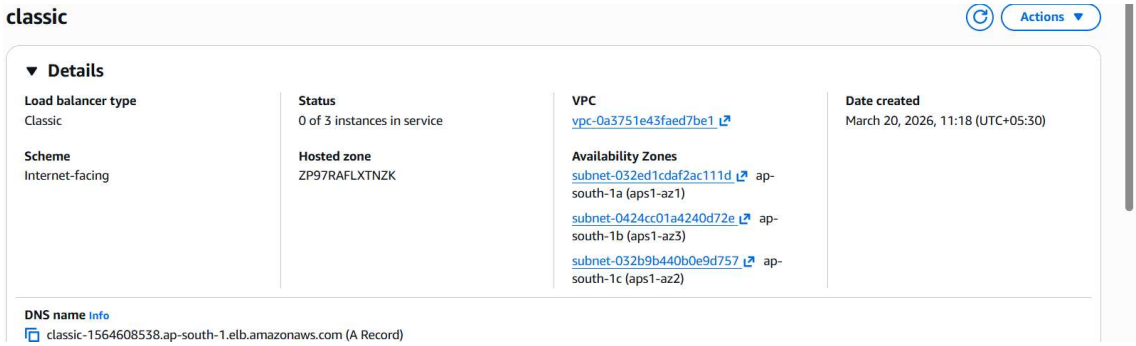




Then we have to add instance to the Load Balancer.



After creation when we hit the load balancer using its DNS will give us output.



Instance – 1

hello from ip-172-31-44-241.ap-south-1.compute.internal

Instance – 2

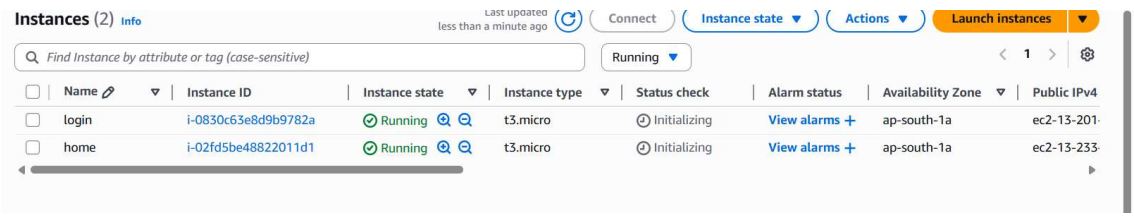
hello from ip-172-31-47-210.ap-south-1.compute.internal

Instance -3

hello from ip-172-31-32-134.ap-south-1.compute.internal

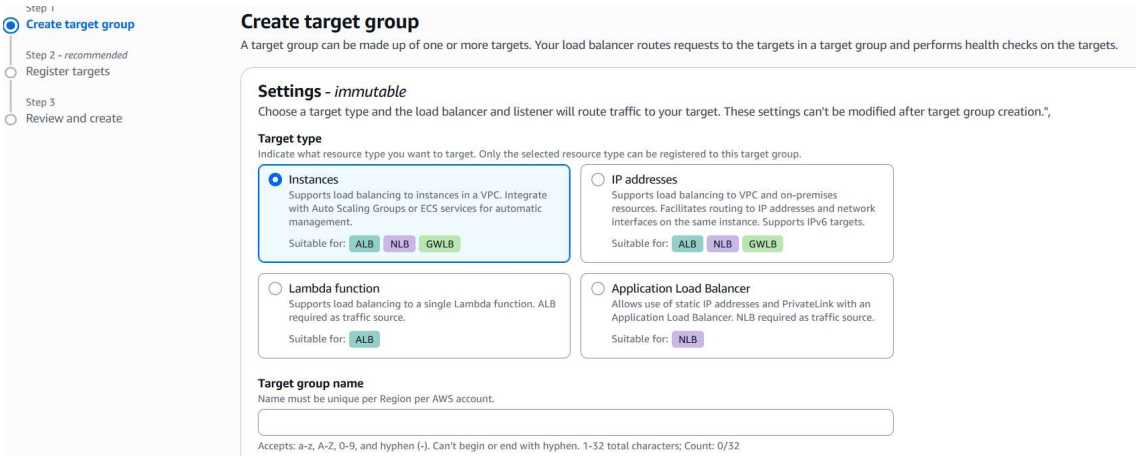
<u>I have created the Application Load balancer with two instances.</u>

The created two instances

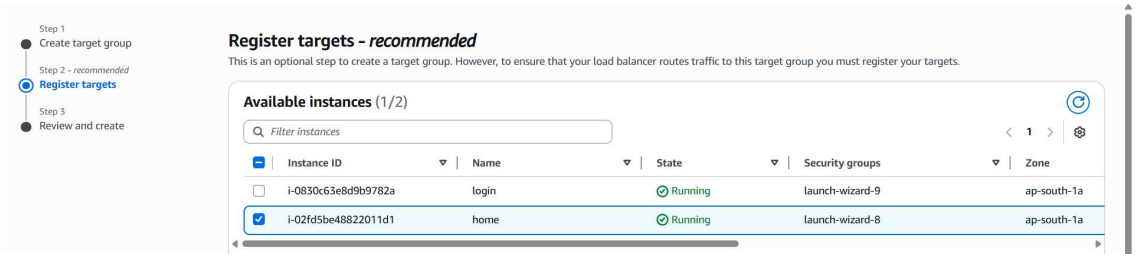


For creating Application Load Balancer we have to Create Target groups for the specified page.

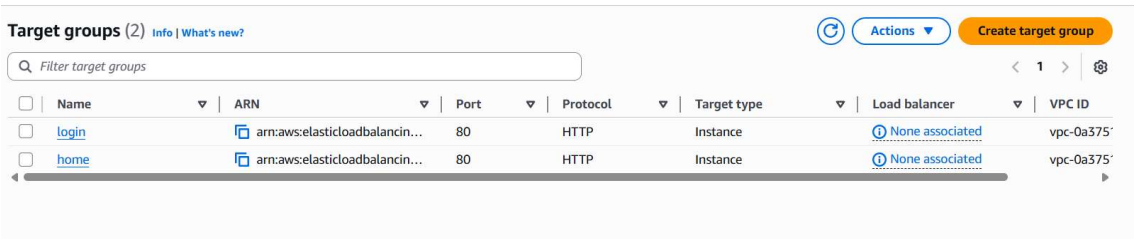
as



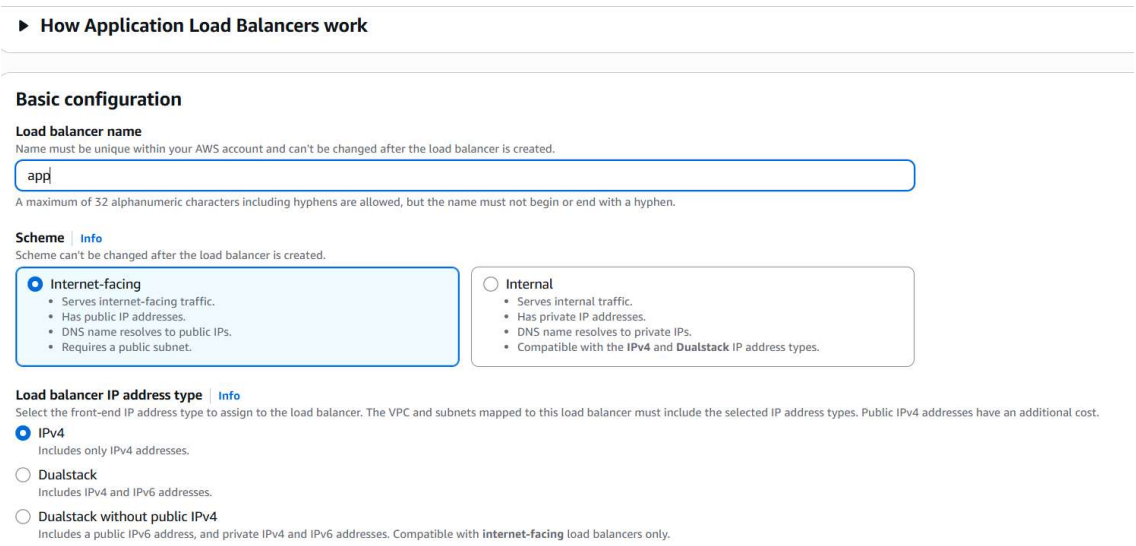
Then we have to assign EC2.



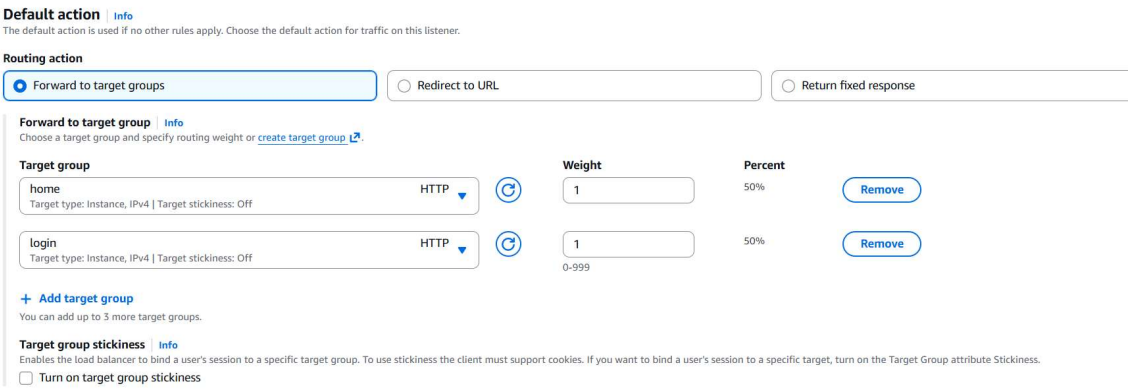
And I have Created Two Target Groups .



Then I have created Application Load Balancer .

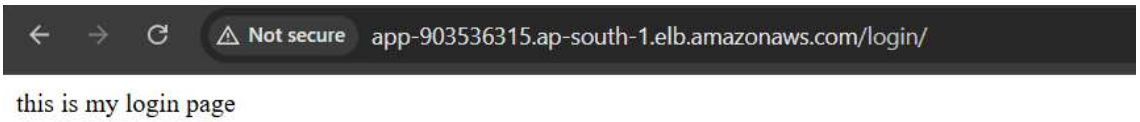


Then I have selected Two Target Groups.

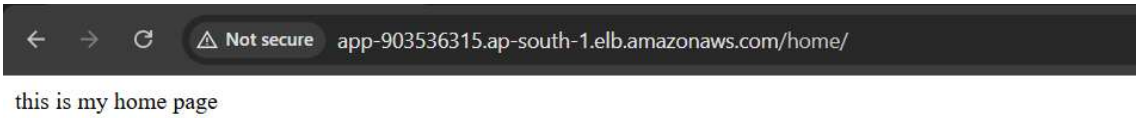


After the creation I have Two Separate page of hosting.

For Login Page.



For Home page



Changing the Httpd Web Server Port Number.

To change the default port in Apache (httpd), you need to modify the main configuration file(s) and then restart the service.

```
[ec2-user@ip-172-31-38-15 ~]$ sudo su
[root@ip-172-31-38-15 ec2-user]# nano /etc/httpd
[root@ip-172-31-38-15 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-38-15 ec2-user]#
```

I have changed the listhening port from 80 to 2118.

```
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts.  See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:2118
Listen 2118

#
# Dynamic Shared Object (DSO) Support
```

After Changing the Port.

hello from ip-172-31-38-15.ap-south-1.compute.internal